Min-Cheol Lee

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Summary

Physicist and research engineer with 7+ years of experience in computational lithography, optical spectroscopy, and scientific computing at Intel and Los Alamos National Laboratory. Develop physics-based models and software for simulation, calibration, and validation using experimental data. Research background in quantum materials, ultrafast optics, and interactions between light and matter.

Experience

Intel Corporation Phoenix, AZ
Software Research Scientist / Engineer, RET/OPC 2022 to Present

- Own OPC models for 20+ EUV and DUV layers on Intel 18A and 14A, including after develop inspection (ADI) and after etch inspection (AEI) models, from calibration through validation and release.
- Introduced ADI and AEI models for transistor layers, reducing defect density by at least a factor of five in both programs. The same work reduced final critical dimension standard deviation by 67% in a 14A re-spin. Received an Intel divisional award for introducing post etch OPC.
- Built Python tools for aerial image analysis, SEM processing, and model calibration. Analyzed 10,000+ SEM images and incorporated fab data into OPC calibration using machine learning.
- Validate models against SEM measurements and work with lithography and integration teams to correlate results with cross sectional TEM, electrical tests, device performance, and optical defect data.
- Collaborate with OPC recipe engineers on full chip correction and validation, refining model parameters and recipes to resolve patterning and verification issues.
- Automated mask geometry checks with Python, Bash, and Siemens Calibre DRC to detect slivers, malformed polygons, and unexpected fragments, shortening qualification from one week to two days.

Los Alamos National Laboratory Los Alamos, NM
Postdoctoral Researcher 2019 to 2022

- Built and aligned ultrafast optical, terahertz, and X-ray experiments to study carrier and lattice dynamics in superconducting and topological quantum materials.
- Developed Python and MATLAB pipelines for instrument control, calibration, signal processing, numerical modeling, and parameter extraction from terabyte-scale experimental datasets.
- Diagnosed alignment drift, signal instability, and detector response issues in synchronized optical experiments. Coauthored research in *Nature Materials*, *Nature Communications*, and *Physical Review Letters*.

Selected Project

litopc | Personal computational lithography project litopc.com

- Developed an interactive optical lithography simulator for the web, combining numerical diffraction and imaging calculations with a visual interface for exploring pattern transfer.

Technical Skills

Programming and analysis: Python, NumPy, SciPy, Pandas, C++, MATLAB, Bash, Linux, Git, numerical simulation, signal processing, model calibration, experimental design.

Physics and engineering: Computational lithography, OPC/RET, EUV/DUV, SEM analysis, ultrafast optical and terahertz spectroscopy, X-ray measurements, experiment automation, quantum materials.

EDA platforms: ASML Tachyon, Siemens Calibre DRC, Intel Mazama (internal OPC platform).

Education

Seoul National University, Seoul, South Korea
Ph.D., Physics (Condensed Matter and Optical Physics) 2019
B.S., Physics 2013

25+ peer-reviewed publications across graduate and postdoctoral research. Google Scholar